

ABDUL RAFAY IMRAN

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PROFESSIONAL SUMMARY

Data Science professional and Software Engineering student with hands-on experience in machine learning, predictive modeling, NLP, and Explainable AI (XAI). Proven ability to deliver end-to-end data science pipelines - from exploratory data analysis and feature engineering through model training, cross-validation, and stakeholder reporting. Skilled in Python, Scikit-Learn, XGBoost, and PyTorch, with a strong foundation in statistical analysis, supervised learning, and production-grade backend APIs.

TECHNICAL SKILLS

- **Core Competencies:** Machine Learning, Predictive Modeling, Feature Engineering, EDA, Statistical Analysis, NLP, Explainable AI (XAI), Supervised Learning, Data Wrangling, ETL Pipelines
- **ML Libraries:** Scikit-Learn, XGBoost, Random Forest, Pandas, NumPy, PyTorch, HuggingFace Transformers, Matplotlib, Seaborn
- **Programming Languages:** Python (Expert), SQL, R, JavaScript, TypeScript
- **Backend & APIs:** FastAPI, Node.js, Express.js, REST APIs, GraphQL, WebSockets
- **Databases & DevOps:** MongoDB, MySQL, Redis, Docker, Git, GitHub, Linux, CI/CD Pipelines

PROFESSIONAL EXPERIENCE

CodeAlpha

Dec 2025 - Jan 2026

Data Science Intern

Remote

- Designed a structured end-to-end pipeline covering EDA, outlier filtering, targeted interpolation, and multicollinearity checks using Python (Pandas, Scikit-Learn), resolving data quality issues that were degrading model accuracy.
- Implemented k-fold cross-validation architectures across all submitted projects to measure model generalizability and ensure production-grade reliability.
- Delivered consistently documented Jupyter notebooks with business metrics and feature importance reporting, reducing stakeholder review cycles by enabling non-technical evaluation of model reliability.
- Built robust supervised learning models (regression and classification) with structured performance summaries - enabling stakeholders to make data-driven decisions without requiring a technical background.

PROJECTS

Reality Check - Explainable AI (XAI) Content Analyzer github.com/RafayImraan/realitycheck

- Built an NLP-powered text analysis platform performing 5+ real-time profiling checks (polarization detection, fear appeals, authority misuse) with a live explainability layer showing probability weight distributions and token-level claim validation.
- Conducted safety and fairness auditing across adversarial inputs to confirm model reliability; delivered interpretable, structured outputs enabling users to critically evaluate media content.

GridMind AI - Infrastructure Failure Predictor github.com/RafayImraan/GridMind

- Engineered a multi-domain hybrid predictive engine combining ML anomaly scoring with cascading failure rules across power, water, and traffic systems, correlated with live weather data.
- Outputs include confidence parameters for each risk state prediction, providing actionable real-time operational reports to support infrastructure decision-making.

Crop Yield Prediction - Pakistan github.com/RafayImraan/crop-yield-prediction-in-pakistan

- Compiled historical soil attributes, climate data, and regional output records to train localized predictive yield matrices using supervised regression models; applied target variable encoding and iterative feature pruning.
- Identified dominant localized factors driving national crop returns, enabling data-backed agricultural planning recommendations.

FrostByte AI - Multi-Layer Cybersecurity Platform github.com/RafayImraan/frostbyte

- Built a hybrid malicious-element detector combining real-time feature-driven ML classifiers with heuristic risk indicators, wrapped in an interactive browser extension.
- Minimized false-alarm rates via custom thresholding logic, improving analyst efficiency through a live monitoring feed.

EDUCATION & CERTIFICATIONS

University of Karachi

B.S., Software Engineering

May 2027

Karachi, Pakistan

Aptech Learning

Diploma, Software Engineering

2022 - Oct 2025

Karachi, Pakistan

CERTIFICATION

- Data Science 101: IBM Skills Network / Cognitive Class (June 2026)
- Kaggle: Intermediate Machine Learning
- Machine Learning Explainability
- Data Visualization